

House Price Predictor

California Housing Dataset — Linear Regression
AI & Machine Learning Internship — Task 1
Maincrafts Technology.

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Internship Domain:	Artificial Intelligence & Machine Learning
Task:	Task 1 — Build & Evaluate a Linear Regression Model

1. Introduction

This report presents the development and evaluation of a Linear Regression model to predict median house prices in California. The California Housing dataset was used, which contains 20,640 records of block-level housing data collected from the 1990 Census. The objective was to follow the complete ML workflow: data loading, exploratory data analysis (EDA), preprocessing, model training, and evaluation.

2. Dataset Overview

The dataset contains 8 input features and 1 target variable (MedHouseVal). All 20,640 records were complete with no missing values, making preprocessing straightforward.

Feature	Description
MedInc	Median income of households in the block
HouseAge	Median age of houses in the block
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Total population of the block
AveOccup	Average number of occupants per household
Latitude	Geographic latitude of the block
Longitude	Geographic longitude of the block
MedHouseVal (Target)	Median house value in \$100,000s

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the data distribution, identify outliers, and analyze feature relationships before model training.

3.1 Statistical Summary

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704	2.068558
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532	1.153956
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-124.350000	0.149990
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-121.800000	1.196000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000	1.797000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-118.010000	2.647250
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-114.310000	5.000010

Figure 1: Statistical summary of all features using df.describe()

Key observations from the statistical summary : MedInc ranges from 0.5 to 15 with a mean of 3.87, indicating wide income variation. MedHouseVal (target) ranges from 0.15 to 5.0 with a mean of 2.07 (\$207,000). Population shows high standard deviation (1132), confirming significant variation across blocks. No missing values were found across all 20,640 records.

3.2 Correlation Analysis

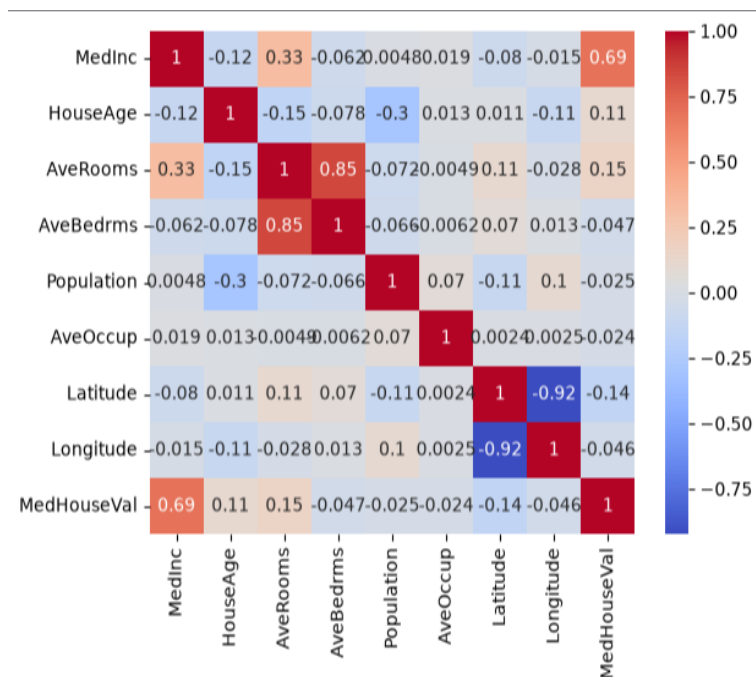


Figure 2: Correlation heatmap of all features

The correlation heatmap reveals the following key insights:

- MedInc has the strongest positive correlation with MedHouseVal (0.69), confirming that income is the primary driver of house prices.
- Latitude and Longitude show strong negative correlation with each other (-0.92), reflecting California's geographic layout.
- AveRooms and AveBedrms are highly correlated (0.85), indicating multicollinearity between these features.
- Most other features show weak correlations with the target, suggesting linear regression may have limitations.

4. Model Training

A Linear Regression model was trained using scikit-learn. The dataset was split into 80% training (16,512 samples) and 20% testing (4,128 samples) using `train_test_split` with `random_state=42` for reproducibility.

Parameter	Value
Algorithm	Linear Regression
Training samples	16,512 (80%)
Testing samples	4,128 (20%)
Random state	42
Features used	All 8 features
Target variable	MedHouseVal

5. Model Evaluation

5.1 Metrics

```
#calculate metrics
mae=mean_absolute_error(y_test, y_pred)
rmse = mean_squared_error(y_test, y_pred)**0.5
r2=r2_score(y_test,y_pred)

print(f"MAE:{mae:.3f}")
print(f"RMSE:{rmse:.3f}")
print(f"R2 SCORE:{r2:.3f}")
✓ 0.0s
MAE:0.533
RMSE:0.746
R2 SCORE:0.576
```

Figure 3: Model evaluation metrics

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.533	On average, predictions are off by \$53,300
RMSE (Root Mean Squared Error)	0.746	Larger errors are penalized; \$74,600 average when include big misses
R ² Score	0.576	Model explains 57.6% of variance in house prices

5.2 Actual vs Predicted Plot

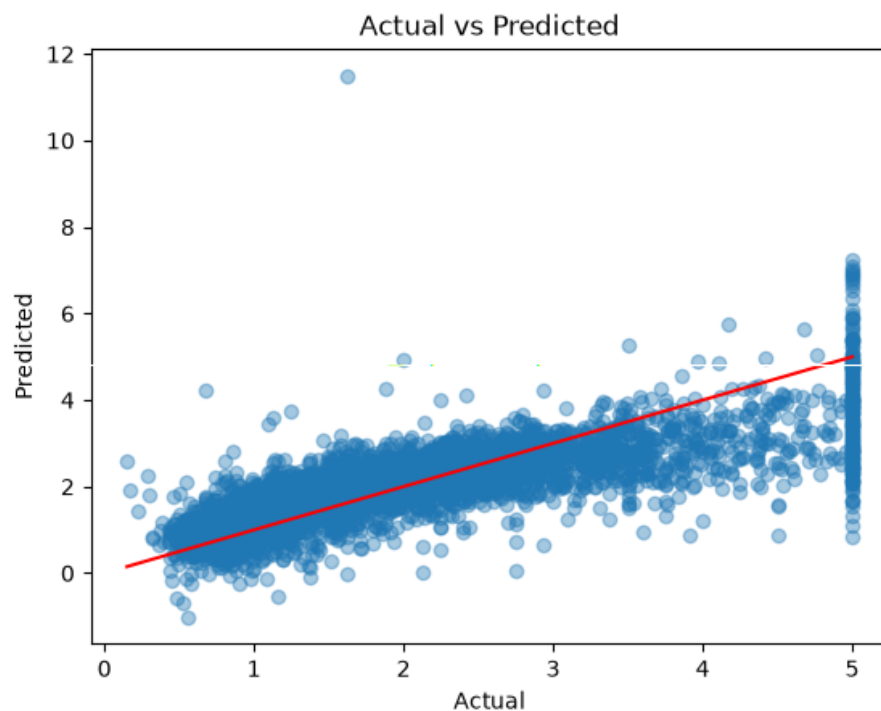


Figure 4: Scatter plot of Actual vs Predicted house prices

The scatter plot shows predictions against actual values with a reference line representing perfect predictions. Key observations: the model performs well for average-priced houses (values 1-3), where dots cluster near the red line. For expensive houses (values 4-5), predictions are more scattered, indicating the linear model struggles with high-value properties. One significant outlier is visible around predicted value 12, likely caused by an extreme data point.

6. Improvement Ideas

Random Forest Regressor: Ensemble of decision trees that can capture non-linear relationships, expected to push R^2 above 0.80.

XGBoost / LightGBM: Gradient boosting algorithms used by Kaggle competition winners; highly effective for tabular data.

Feature Engineering: Creating new features like rooms-per-person ratio or income-per-room could improve model accuracy.

Outlier Removal: Removing extreme values (like the outlier visible in the scatter plot) could reduce RMSE significantly.

Ridge / Lasso Regression: Regularized versions of linear regression that handle multicollinearity better (AveRooms/AveBedrms issue).

Cross-Validation: Using k-fold cross-validation instead of a single train-test split gives a more reliable performance estimate.

7. Conclusion

A Linear Regression model was successfully built and evaluated on the California Housing dataset. The model achieved an R^2 score of 0.576, explaining 57.6% of the variance in house prices. MedInc (median income) was identified as the strongest predictor with a correlation of 0.69. While the baseline model provides reasonable predictions for average-priced homes, it struggles with high-value properties due to the limitations of linear assumptions. Future work should explore ensemble methods like Random Forest or XGBoost to improve predictive accuracy.